

$$SX = \text{[Diagram of a sphere with horizontal lines and blue arcs]} = CX \sqcup_x CX = \frac{X \times [-1, 1]}{\left\{ \begin{array}{l} (a, 1) \sim (b, 1) \\ (a, -1) \sim (b, -1) \end{array} \right\}}$$

$$SY = \text{[Diagram of a sphere with a central circle and blue arcs]} = CY \sqcup_Y CY = S^2!$$